

WEEK 3 — ADVANCED CYBERSECURITY PRACTICAL ASSESSMENT

Network Security Assessment, Traffic Analysis, Vulnerability Assessment & Security Hardening

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1. Introduction

This report documents Week 3 of the DG Interns Hub cybersecurity internship track: a hands-on network security assessment performed entirely within an authorized, isolated VirtualBox lab. The work covers host discovery, port and service enumeration, OS fingerprinting, vulnerability scanning with Nmap, packet-level traffic analysis with Wireshark, correlation between scan activity and captured traffic, a basic vulnerability/risk assessment, and security hardening with before/after verification.

All findings in this report are drawn directly from the intern's own command output and screenshots. Where the available evidence was incomplete or inconclusive for a particular sub-task, this is stated explicitly rather than filled in with assumed or example results, in line with the accuracy requirements of this assessment.

All scanning, vulnerability testing, and traffic analysis described in this report were performed exclusively against an authorized local VirtualBox lab environment (Windows 11 host and an Ubuntu Desktop virtual machine on a Host-Only network). No public or third-party systems were scanned or tested.

2. Objectives

The objective of Week 3 was to build practical, real-world security assessment skills by working through the full lifecycle a security analyst follows when assessing a network: discover, enumerate, capture, analyse, assess, harden, and verify. Specifically, the practical work involved:

- Network and host discovery within an authorized lab segment
- Port scanning, service enumeration, and service/version detection
- Operating system fingerprinting
- TCP SYN scanning and limited UDP scanning
- Investigating Nmap's NSE vulnerability scripts
- Capturing and analysing live network traffic with Wireshark (DNS, TCP, UDP, ICMP, ARP)
- Correlating Nmap scan activity with packet-level Wireshark evidence
- Identifying and risk-classifying security findings
- Implementing security hardening measures and verifying their effect with before/after testing

3. Lab Environment

Host System

- Windows 11 (build 10.0.22000.2538)
- Nmap 7.991 installed on Windows
- Wireshark installed on Windows
- Oracle VirtualBox installed

Virtual Machine (Authorized Target)

- Ubuntu Desktop VM
- IP address: 192.168.56.101 (interface enp0s3)
- Network: 192.168.56.0/24 (VirtualBox Host-Only adapter)
- VM network interface MAC address: (Oracle VirtualBox virtual NIC)

The Ubuntu VM's network configuration was verified directly on the VM using `ip addr`, `hostname -I`, and `ip route`, confirming the 192.168.56.101 address on the `enp0s3` interface.

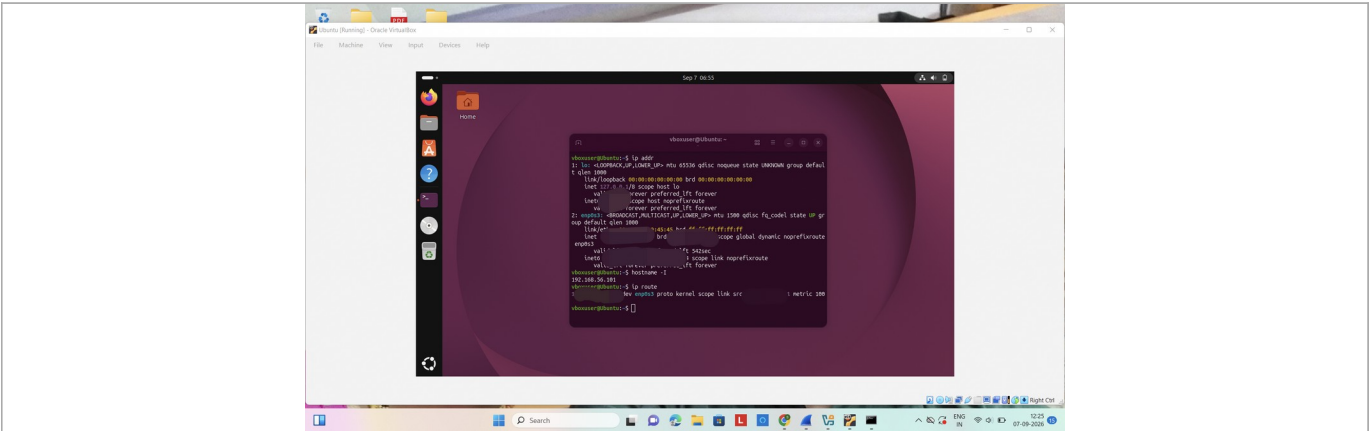


Figure 1. Ubuntu lab VM network configuration — `ip addr / hostname -I / ip route` confirming 192.168.56.101 on `enp0s3`.

Only the Windows host and this single Ubuntu VM were used as the practical target for scanning and traffic analysis; no additional virtual machines were used in the assessment.

4. Network Diagram

The lab topology consists of the Windows 11 host connected to an isolated VirtualBox Host-Only network (192.168.56.0/24), with the Ubuntu Desktop VM as the sole authorized target on that segment. Nmap scans were issued from the Windows host toward the Ubuntu VM; Wireshark captured traffic on the Windows host's active network interface.

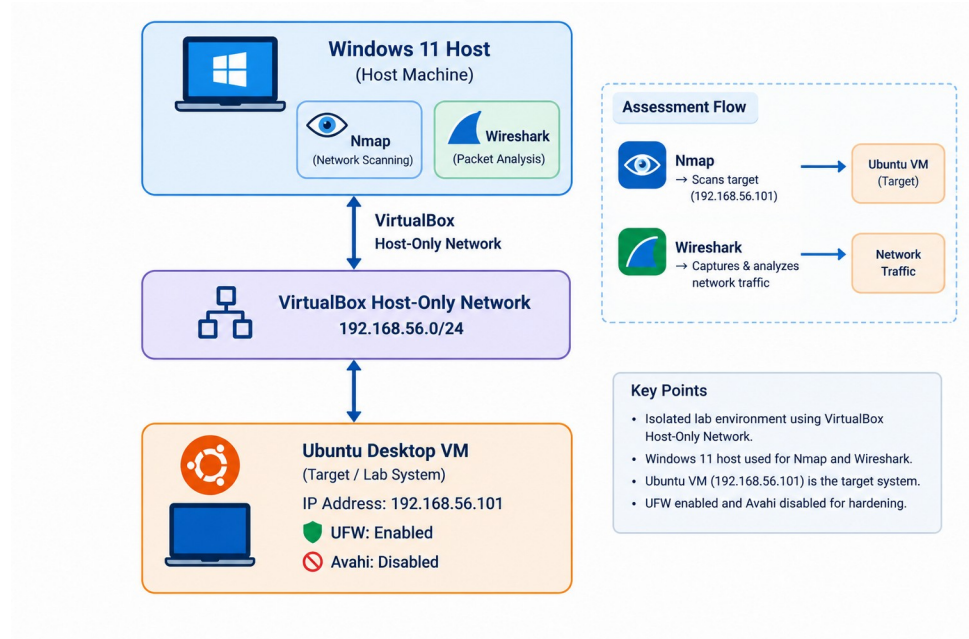


Figure 2. Authorized VirtualBox Host-Only cybersecurity assessment topology and assessment flow.

5. Tools Used

Tool	Purpose
Windows 11	Host operating system used to run Nmap and Wireshark
Ubuntu Desktop (VM)	Authorized target virtual machine
Oracle VirtualBox	Virtualization and isolated Host-Only lab networking
Nmap 7.991	Network discovery, port scanning, service/version enumeration, OS detection, and NSE vulnerability scripts
Wireshark	Packet capture and network traffic analysis
Network diagram (draw.io style)	Network topology illustration (Figure 2)
GitHub	Project documentation and submission

6. Task 7 — Network Discovery & Basic Nmap Scanning

Objective: understand how a security analyst discovers authorized devices and exposed network services on a segment before assessing them individually.

Host Discovery

Command: `nmap -sn 192.168.56.0/24`

```
nmap -sn 192.168.56.0/24
```

The scan swept the full /24 range (256 addresses) and identified three active hosts in 5.47 seconds:

IP Address	MAC Address	Role
192.168.56.1	Not reported	VirtualBox Host-Only adapter (gateway)
192.168.56.100	08:00:27:1A:98:F6 (Oracle VirtualBox)	Another VirtualBox virtual adapter/VM on the segment
192.168.56.101	(Oracle VirtualBox)	Ubuntu Desktop VM — the authorized target

Basic Nmap Scan

Command: `nmap 192.168.56.101`

```
nmap 192.168.56.101
```

The default scan (top 1,000 TCP ports) reported the host up (0.0012s latency) with all 1,000 scanned ports in a closed state (reset), meaning no open TCP ports were found on the default port set.

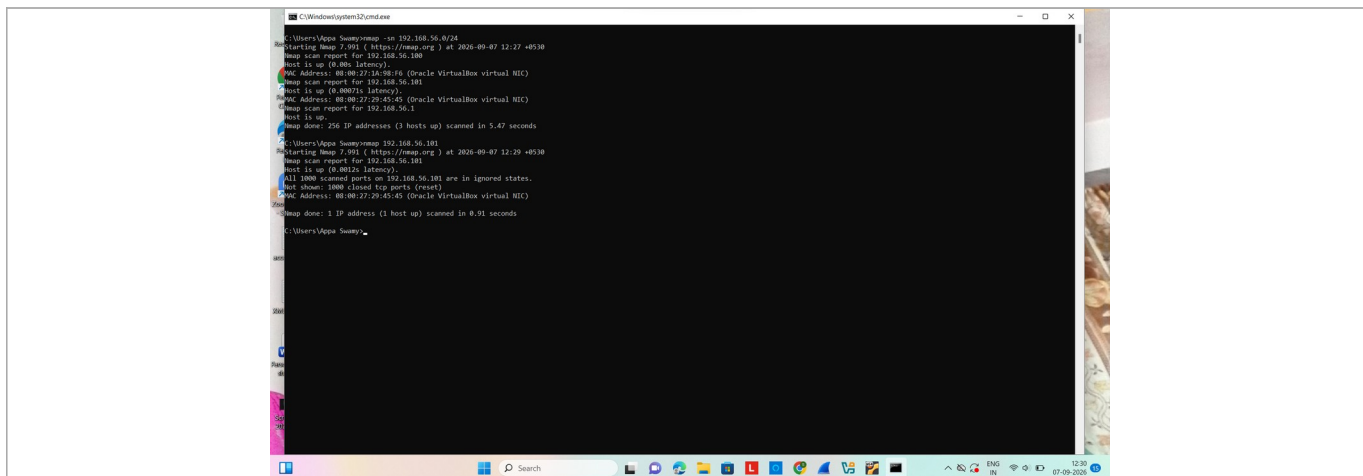


Figure 3. Nmap host discovery (`nmap -sn`) and basic scan (`nmap 192.168.56.101`) — all 1,000 default ports closed.

Accuracy note: the basic scan found no open TCP ports on 192.168.56.101. This report does not claim any specific service (e.g. SSH, HTTP) was found open on the default scan, since none was observed in the evidence.

7. Task 8 — Advanced Nmap Security Assessment

The following scans were performed against the authorized target, 192.168.56.101.

7.1 Service/Version Detection & 7.2 OS Detection

Commands: `nmap -sV 192.168.56.101` / `nmap -O 192.168.56.101`

Service detection (-sV) found no open ports to fingerprint — the result matched the default scan, with all 1,000 ports closed. OS detection (-O) explicitly warned that results "may be unreliable" without at least one open and one closed reference port; it returned a list of unrelated "aggressive OS guesses" (routers, IP cameras, a switch, cable modems) at 96% confidence but concluded "No exact OS matches for host (test conditions non-ideal)." Neither result should be treated as a confirmed finding.

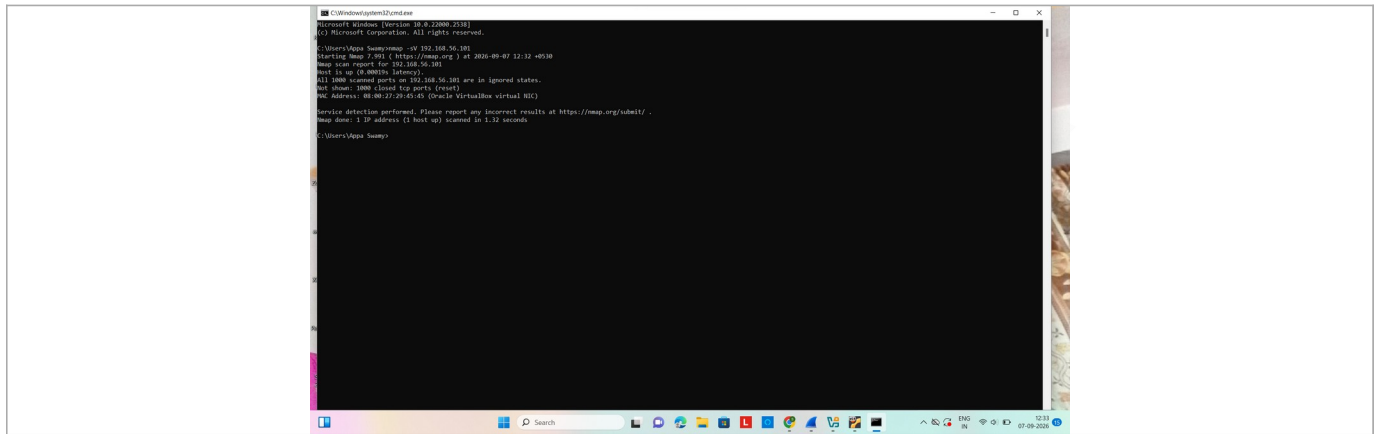


Figure 4. `nmap -sV` — no open ports to fingerprint.

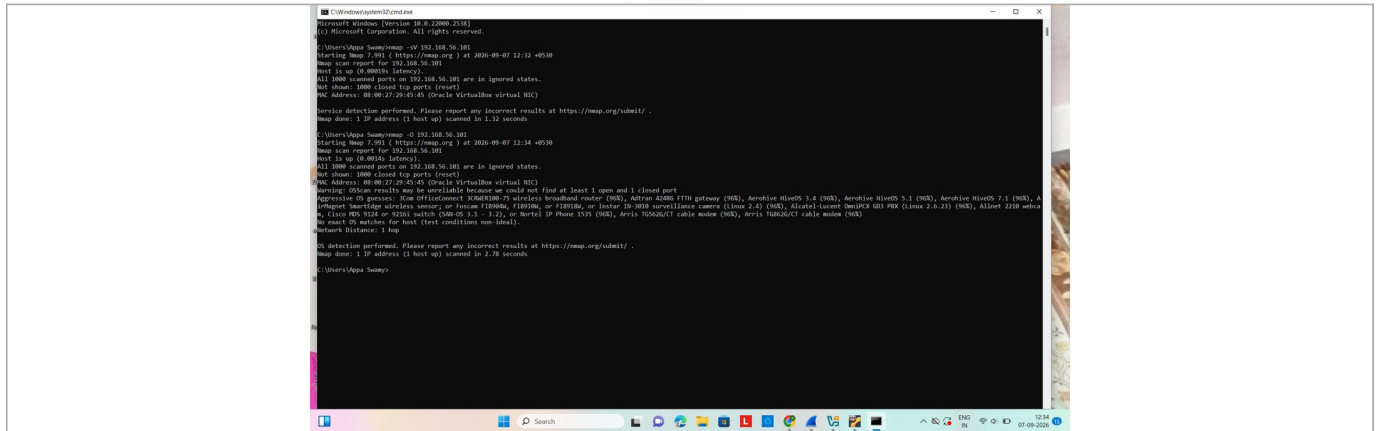


Figure 5. `nmap -O` — inconclusive, no exact OS match.

7.3 TCP SYN Scan & 7.4 UDP Top-20 Scan

Commands: `nmap -sS 192.168.56.101` / `nmap -sU --top-ports 20 192.168.56.101`

The SYN scan (-sS) confirmed the same result as the default scan: all 1,000 TCP ports closed (reset). The UDP top-20 scan checked the 20 most common UDP ports (53, 67–69, 123, 135, 137–139, 161, 162, 445, 500, 514, 520, 631, 1434, 1900, 4500, 49152) — all reported closed. Port 5353 (mDNS/Avahi) is not part of this top-20 list and was investigated separately (Section 12).

8. Nmap Results Summary & Advanced Analysis

Consolidated Results Table

The table below distinguishes clearly between confirmed closed ports, the ambiguous open|filtered UDP state, and services confirmed only at the host level (via commands run on the Ubuntu VM itself, as reported by the intern) — these are not the same as a confirmed Nmap-open port.

Target	Port(s)	Protocol	Nmap State	Service (if known)	Notes
192.168.56.101	1–1000 (default set)	TCP	Closed (reset)	—	No open TCP ports found on the default scan, SYN scan, or -sV scan
192.168.56.101	Top-20 common ports	UDP	Closed	—	No open UDP ports among the 20 most common
192.168.56.101	53, 631, 5353	TCP	Closed	domain / ipp / mdns (Nmap labels)	Consistent with DNS resolver, CUPS, and Avahi binding to loopback only (see Section 14)
192.168.56.101	5353	UDP	open filtered	zeroconf (mDNS/Avahi)	Nmap cannot confirm truly open vs. filtered from a remote UDP probe; see Finding 1

The open|filtered state on UDP 5353 is NOT the same as a confirmed open port. It means Nmap received no response and cannot distinguish between "open with no reply" and "filtered/dropped by a firewall." This state persisted even after hardening (Section 13.2), which is expected Nmap UDP-scanning behaviour rather than evidence that Avahi was still running — host-level checks (ss, systemctl) are the authoritative source for that.

Advanced Analysis

Default TCP/UDP port set — no open services observed

What service is running: None was confirmed open on the scanned TCP (1–1000) or top-20 UDP ports

Why it matters: A clean result on the standard scanned ports is a good baseline sign — it means no common, easily discoverable services were remotely reachable at scan time

Is exposure necessary: N/A — no exposure was found on these ports

Risk if it were vulnerable: N/A

How to keep it this way: Maintain the host firewall (UFW), avoid installing/enabling network-facing services unless required, and re-scan periodically to confirm no new ports have opened

UDP 5353 — mDNS/Avahi (open|filtered)

What service is running: Avahi (mDNS/zeroconf), confirmed at the host level via ss output as listening on UDP 0.0.0.0:5353 and [::]:5353 before hardening

Why it might be required: mDNS/Avahi enables automatic local network service discovery (e.g. finding printers or other devices on a LAN) without manual configuration

Is network exposure necessary: Not for a general-purpose desktop/lab VM with no local discovery use case — this is why it was selected for hardening

Risk if vulnerable: mDNS implementations have historically been associated with information disclosure and amplification-style risks on some platforms; leaving it enabled increases attack surface without a clear benefit in this lab context

How it was secured: Avahi was disabled (Section 13.2), removing the service from startup and confirmed inactive via systemctl

Loopback-only services (DNS resolver, CUPS)

Per the intern's reported Ubuntu ss output, systemd-resolved (DNS, ports 53/UDP) was bound to 127.0.0.54 and 127.0.0.53, and CUPS (port 631/TCP) was bound to 127.0.0.1 and [::1] — i.e. loopback-only, not exposed to the network. This is corroborated by the external scan in Figure 8, which showed TCP port 631 as closed when probed remotely, consistent with a loopback-only binding rather than external exposure. Per the accuracy rules for this assessment, these are not described as externally exposed services.

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9. Task 9 — Wireshark Network Traffic Capture

Wireshark was used on the Windows host to capture and analyse authorized network traffic. Two capture sessions were recorded on the host's active "Wi-Fi" interface: a primary session of approximately 3,105 packets (ARP, DNS-query, TCP SYN, TCP RST, and general traffic views below) and a shorter secondary session of approximately 2,567 packets used for supplementary DNS analysis.

Scope note: these Wireshark captures were taken on the Windows host's general network interface, not on the isolated VirtualBox Host-Only segment used for the Nmap scans. They demonstrate normal DNS/TCP/UDP/ICMP/ARP traffic types as required by the task, but do not by themselves show traffic to or from 192.168.56.101 — this distinction is carried through into Task 10.

DNS Traffic & TCP Traffic (RST)

DNS (filter `dns.flags.response == 0`): queries observed for `accounts.google.com`, `oauthaccountmanager.googleapis.com`, and `update.googleapis.com` — normal background browser/OS traffic. TCP RST (filter `tcp.flags.reset == 1`): two RST packets on port 443 (59246→443 and 59245→443, IPv6 source 2405:201:c052:118f:::5f09) — routine TCP connection teardown, not indicative of malicious activity.

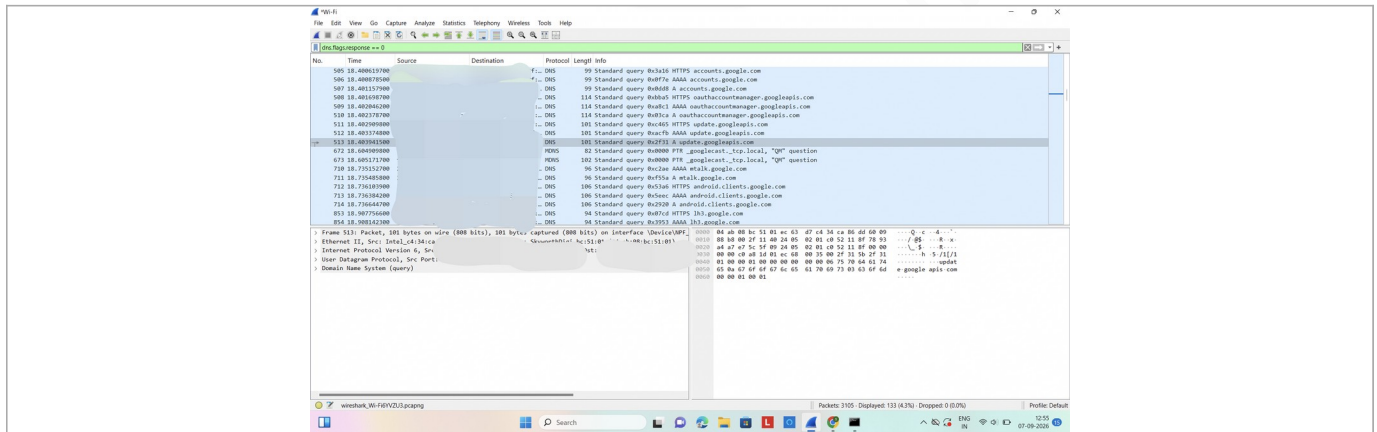


Figure 9. Wireshark — DNS queries only (`dns.flags.response == 0`).

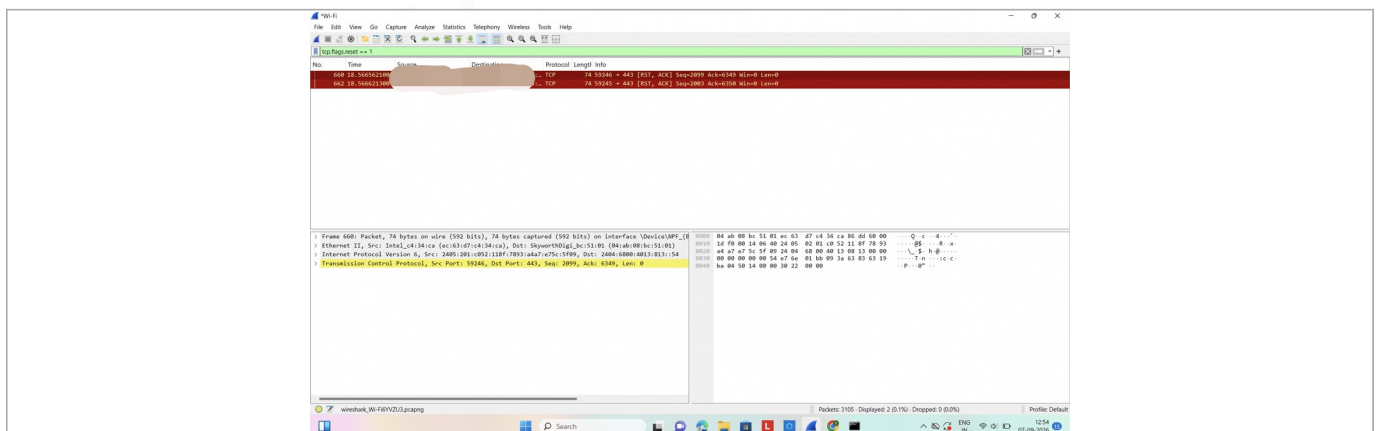


Figure 10. Wireshark — TCP RST packets on port 443.

TCP Traffic (SYN) & General Traffic (Unfiltered)

TCP SYN (filter `tcp.flags.syn == 1`): numerous SYN/SYN-ACK packets on port 443 with browser-typical window sizes and MSS values — consistent with a browser opening several parallel HTTPS connections, not scan activity. Unfiltered view:

a mix of QUIC "Protected Payload" packets and ICMPv6 Neighbor Solicitation/Advertisement for 2405:201:c052:118f::c0a8:1d01 — normal modern browser/OS network behaviour.

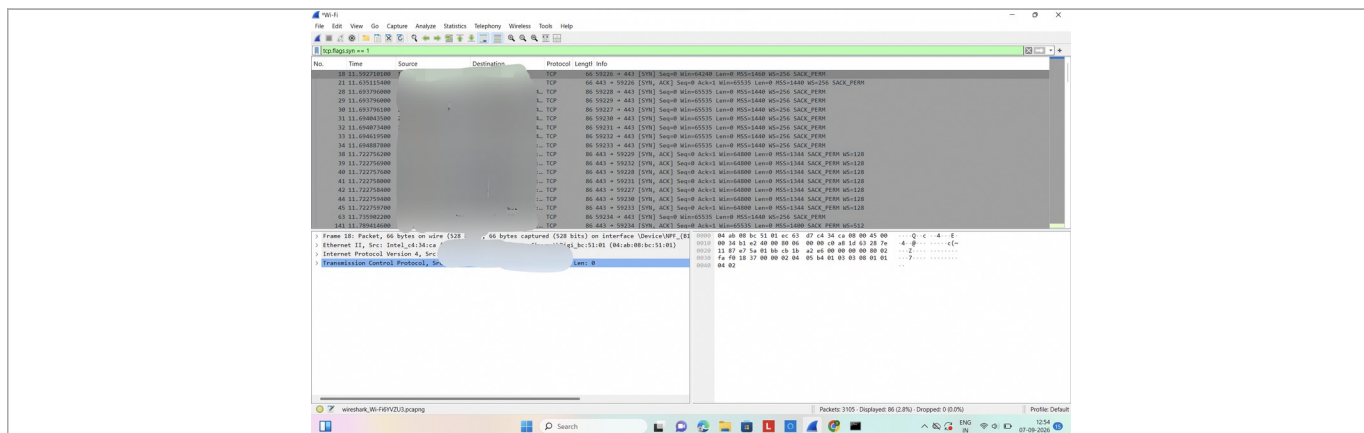


Figure 11. Wireshark — TCP SYN packets, normal HTTPS handshakes.

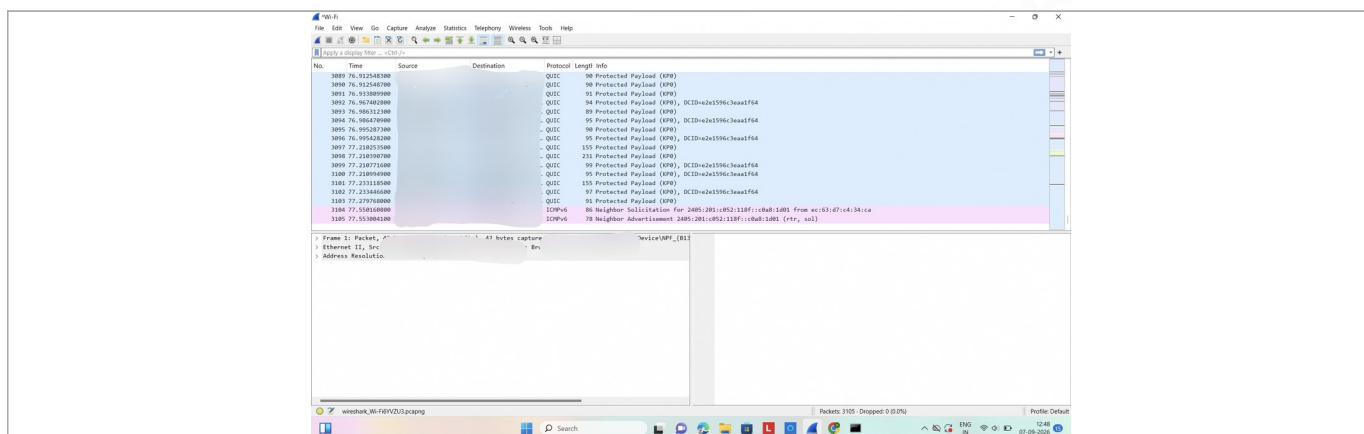


Figure 12. Wireshark — unfiltered capture: QUIC + ICMPv6.

UDP Traffic & ARP Traffic

UDP (filter udp): dominated by DNS-over-UDP lookups for content-delivery/telemetry domains (msedge.net, footprintdns.com CNAME chains) — normal Windows/Edge background activity. ARP (filter arp): broadcast requests on the host's regular 192.168.29.0/24 LAN (separate from the 192.168.56.0/24 lab segment), e.g. Intel_39:b9:35 asking "Who has 192.168.29.1? Tell 192.168.29.131" — routine ARP resolution.

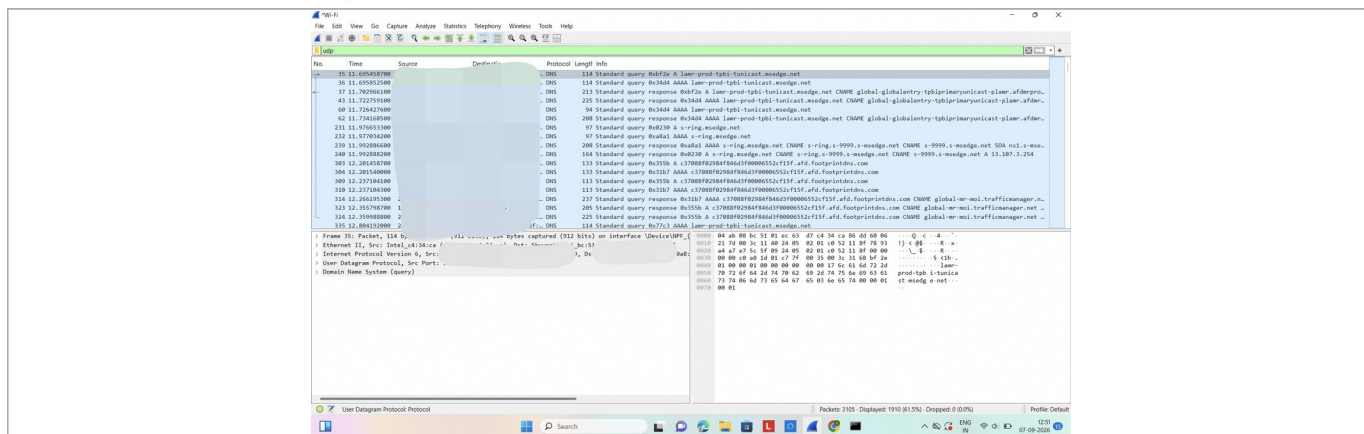


Figure 13. Wireshark — UDP traffic, DNS-over-UDP lookups.

10. Task 10 — Nmap + Wireshark Investigation

Objective: understand what an Nmap scan looks like at the packet level by correlating Nmap output with simultaneously captured Wireshark evidence.

Command used for the scan: `nmap -sS 192.168.56.101`

Evidence gap: the Wireshark screenshots available for this report (Task 9, Figures 9–14) capture general host traffic on the Windows Wi-Fi interface and do not show packets with 192.168.56.101 as the source or destination. No packet-level capture specifically correlating an Nmap SYN scan against the Ubuntu VM was included in the evidence provided. Rather than inventing packet numbers or flags for this correlation, this section documents the expected behaviour pattern and recommends capturing this evidence directly on (or filtered to) the 192.168.56.0/24 segment in a follow-up capture.

Expected Nmap-to-Wireshark Correlation (Reference Pattern)

The table below documents the standard interpretation of SYN-scan results against Wireshark evidence, for use once a correlated capture is available. It is provided as reference methodology, not as a claim that these exact packets were observed against 192.168.56.101.

Nmap Result	Expected Wireshark Evidence	Interpretation
Port detected open	SYN → SYN/ACK	Service appears reachable and listening
Port detected closed	SYN → RST	Port is reachable but not accepting connections
Port filtered / no response	No expected response observed	Possible firewall or filtering in place

Since the actual SYN scan against 192.168.56.101 (Section 7.3, Figure 6) reported all 1,000 ports as closed (reset), the expected packet-level signature — had it been captured on the lab segment — would show a SYN sent by Nmap to each port followed by an RST response from the Ubuntu VM, consistent with the "closed" interpretation row above.

11. Task 11 — Advanced Network Traffic Investigation

This investigation draws on the same Wireshark evidence captured in Task 9, examining it for the most active hosts, protocol breakdown, and any traffic that may warrant a closer look. As instructed, no traffic below is labelled malicious purely on suspicion — each entry is described by what was observed and why (if at all) it may be worth a second look.

Top Communicating Hosts

The most frequently observed source/destination pairs in the primary capture involved the local device (Intel_c4:34:ca / Intel_39:b9:35 network adapters) communicating with Google and Microsoft cloud IPv6 ranges (2404:6800::/32 and 2405:201::/32 prefixes) for DNS, QUIC, and HTTPS traffic — consistent with normal cloud-service usage rather than a small number of dominant, unusual peers.

Investigation Table (5 Observed Events)

Time	Source	Destination	Port/Type	Protocol	Observation	Security Significance
~12:53	Intel_39:b9:35 (local)	Broadcast	—	ARP	"Who has 192.168.29.1? Tell 192.168.29.131"	Normal ARP resolution; no action needed
~12:55	2405:201:c052:118 f:::5f09	2404:6800:4013:81 3::54	59246 → 443	TCP (RST)	Connection reset after HTTPS session	Routine TCP teardown; not suspicious
~12:54	Local host adapter	Multiple :443 peers	59226–	TCP	Multiple parallel HTTPS	Typical of a browser

Time	Source	Destination	Port/ Type	Protocol	Observation	Security Significance
			59234 → 443	(SYN)	handshakes	loading many resources at once
~12:51	Local host adapter	Various IPv6 peers	53 (DNS over UDP)	UDP/ DNS	Queries to *.msedge.net / footprintdns.com CNAME chains	Standard Microsoft/CDN telemetry and content delivery — worth periodic review if unfamiliar domains appear
~14:44	Local host adapter	Google/ Microsoft/Bing IPs	53 (DNS)	DNS	Queries for settings- win.data.microsoft.com, th.bing.com, clientservices.googleapis.co m	Normal OS/browser telemetry; monitor for unexpected additions over time

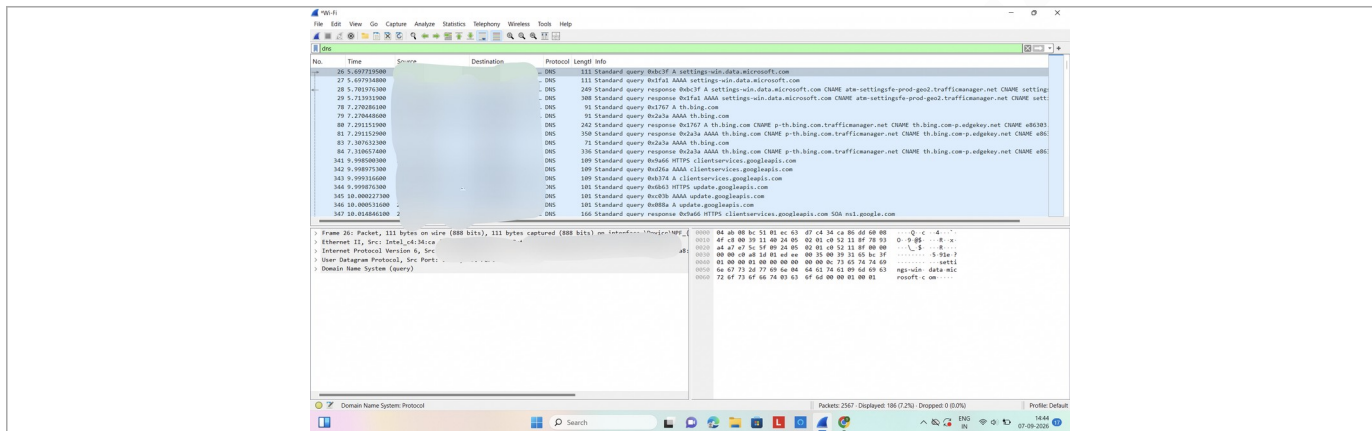


Figure 15. Wireshark — secondary capture session (dns filter) used as supplementary evidence for the traffic investigation.

None of the five events above met the bar for a malicious classification; all are explainable as normal ARP resolution, TCP session teardown, browser connection behaviour, or OS/CDN telemetry. They are recorded here as the investigation's due-diligence review rather than as active incidents.

12. Task 12 — Vulnerability / Security Assessment

Combining the results of Tasks 7–11, five security findings were identified. Each is risk-classified with reasoning, drawing only on confirmed evidence.

Finding 1 — Potential mDNS/Avahi Exposure (UDP 5353)

Risk: Medium

Evidence: Ubuntu ss output (as reported) showed avahi-daemon listening on UDP 0.0.0.0:5353 and [::]:5353. A targeted Nmap scan (nmap -sU -p 5353 -sV 192.168.56.101) reported the port as open|filtered, zeroconf.

Reasoning: Medium rather than High because mDNS is a local-discovery convenience service rather than a directly exploitable remote-code path in this evidence; Medium rather than Low because it is unnecessary on this VM and does increase attack surface

Potential Impact: Unnecessary network-facing service that could expose local network information or be leveraged in local reconnaissance

Recommendation: Disable Avahi/mDNS if local service discovery is not required (implemented — see Section 13.2)

Nmap's open|filtered state is not a confirmed open port — it means no response was received, which could mean the service is open with no reply, or the packet was filtered. This finding relies on the host-level ss evidence, not the Nmap state alone, to confirm the service was actually listening.

Finding 2 — UFW Firewall Initially Inactive

Risk: Medium

Evidence: sudo ufw status verbose returned "Status: inactive" prior to hardening.

Reasoning: Medium because, while the default scan found no open ports even without UFW, an active host firewall is a standard defence-in-depth control; its absence is a process/configuration gap rather than an active exploit path

Potential Impact: Without a host firewall, any future service that is installed and starts listening would be immediately reachable from the network with no additional control in place

Recommendation: Enable UFW and maintain default-deny inbound rules (implemented — see Section 13.1)

Finding 3 — Pending Ubuntu Package Updates

Risk: Medium

Evidence: apt list --upgradable listed nine upgradable packages: gnome-control-center-data, gnome-control-center-faces, gnome-control-center, libaudit-common, libaudit1, libflashrom1, python-apt-common, python3-apt, and xdg-terminal-exec.

Reasoning: Medium because outdated packages can carry known, patched vulnerabilities; none of the listed packages are shown in the evidence to have an active exploited vulnerability, so this is a hygiene finding rather than a confirmed critical exposure

Potential Impact: Running outdated package versions increases exposure to publicly known issues that have already been fixed upstream

Recommendation: Apply operating system and security updates regularly (sudo apt update && sudo apt upgrade -y)

Finding 4 — UDP Scan Ambiguity (open|filtered results)

Risk: Medium

Evidence: The targeted UDP 5353 scan returned open|filtered rather than a definitive open or closed state.

Reasoning: Medium because this represents a genuine blind spot in remote assessment coverage — a Medium-risk service could hide behind this ambiguous state without an external scan being able to confirm it

Potential Impact: Relying on Nmap UDP results alone could under- or over-estimate real exposure; open | filtered ports must not be reported as confirmed open in any assessment

Recommendation: Always validate UDP findings against host-level evidence (e.g. `ss -ltnup`) rather than the Nmap state alone, and restrict unnecessary UDP services at the host firewall

Finding 5 — Unnecessary Services Increase Attack Surface

Risk: Low-Medium

Evidence: Service enumeration (host-level `ss` output) identified Avahi and other locally-running components (`chronyd`, `systemd-resolved`, `cupsd`) alongside the confirmed-unnecessary Avahi service

Reasoning: Low-Medium because most of these services (`chronyd`, `systemd-resolved`, `cupsd`) were confirmed bound to loopback only and pose minimal external risk; the Low-Medium rating reflects the general principle that every additional running service is one more thing to maintain and monitor, not a specific active exposure

Potential Impact: A larger running-service inventory increases the maintenance and patching burden and the chance that a future misconfiguration exposes a service unintentionally

Recommendation: Periodically review running services and listening ports; disable or remove anything not required for the system's actual purpose

Findings Summary Table

#	Finding	Evidence	Risk	Recommendation
1	Potential mDNS/Avahi exposure (UDP 5353)	<code>ss</code> output + targeted Nmap scan (open filtered)	Medium	Disable Avahi/mDNS if not required
2	UFW firewall initially inactive	<code>sudo ufw status verbose</code> → inactive	Medium	Enable UFW, maintain firewall rules
3	Pending Ubuntu package updates	<code>apt list --upgradable</code> (9 packages)	Medium	Apply updates regularly
4	UDP scan ambiguity (open filtered)	Targeted UDP 5353 scan result	Medium	Validate with host-level evidence (<code>ss</code>)
5	Unnecessary services increase attack surface	Service enumeration via <code>ss</code>	Low-Medium	Review and minimise running services

13. Task 13 — Security Hardening

Three security improvements were implemented on the Ubuntu VM in response to the findings above.

Improvement 1 — Enable UFW

Before: UFW status was inactive

Action: `sudo ufw enable`

After: UFW became active, with a default-deny-incoming / allow-outgoing policy, and enabled at system startup

```
sudo ufw status verbose # Before: Status: inactive
sudo ufw enable          # Firewall is active and enabled on system startup
sudo ufw status verbose # After: Status: active
```

Improvement 2 — Disable Avahi/mDNS

Before: Avahi was listening on UDP 5353 (per host-level ss output)

Action: `sudo systemctl disable --now avahi-daemon`, then `avahi-daemon.socket`

Verification: `systemctl is-active avahi-daemon` → inactive; `systemctl is-active avahi-daemon.socket` → inactive

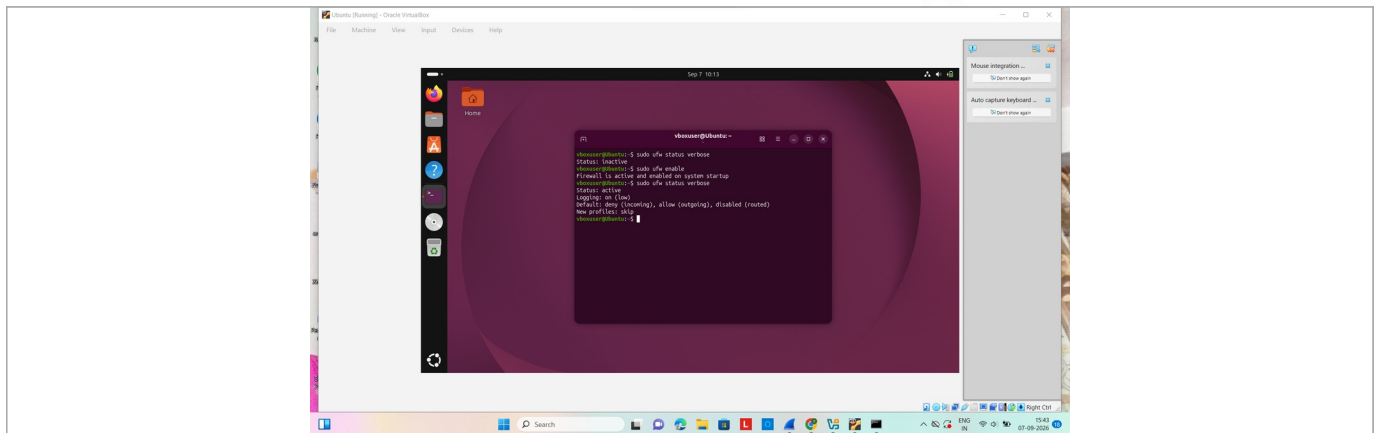


Figure 16. UFW status before (inactive) and after (active) enabling the firewall.

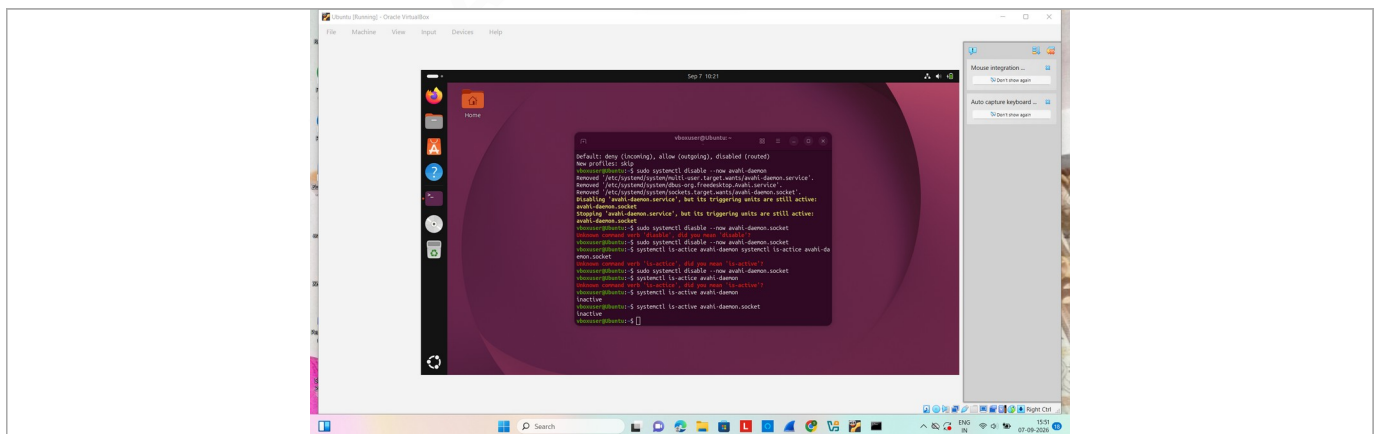


Figure 17. Disabling Avahi — both units confirmed inactive.

Nmap re-check: A follow-up targeted scan (`nmap -sU -p 5353 -sV 192.168.56.101`) still reported 5353/udp as open | filtered — see note below

Improvement 3 — Review Ubuntu Package Updates

Commands: `sudo apt update`, then `sudo apt upgrade -y`

Verification method: apt list --upgradable

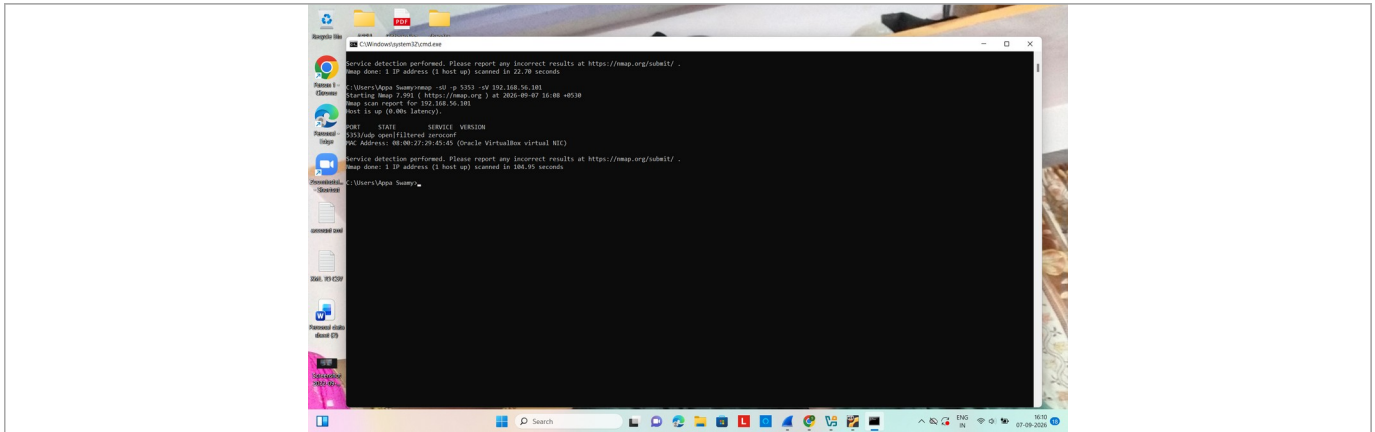


Figure 18. Post-hardening UDP 5353 re-scan — still open|filtered (Nmap limitation, not a failed fix).

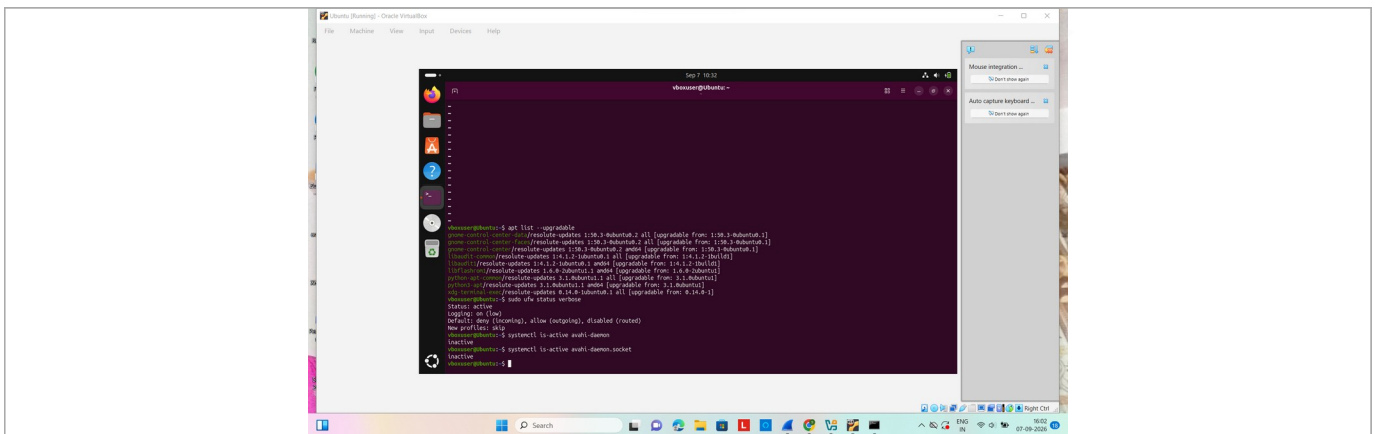


Figure 19. Final verification — pending updates list, UFW active, Avahi inactive.

Figure 18 is an expected result, not a failed fix: Nmap's UDP scanning cannot distinguish "nothing is listening" from "a firewall silently dropped the probe" — both look identical from outside. The host-level systemctl checks (Figure 17 and 19) are the authoritative confirmation that Avahi was successfully disabled.

The evidence captured confirms the list of pending packages and re-confirms UFW active and Avahi inactive at the same time (Figure 19). A separate post-upgrade re-run of `apt list --upgradable` showing an empty/reduced list was not included in the provided evidence; this should be captured as follow-up verification if the upgrade commands were run to completion.

Security benefit: applying updates closes the gap between publicly known, already-patched issues and the software actually running on the system, reducing exposure to well-understood attack techniques.

14. Before / After Comparison

Security Issue	Before	Action Taken	After	Improvement
UFW firewall	Inactive	<code>sudo ufw enable</code>	Active — default deny incoming, allow outgoing, enabled at startup	Adds a host-level barrier against unsolicited inbound

Security Issue	Before	Action Taken	After	Improvement
				connections
Avahi / mDNS	Listening on UDP 5353 (per ss)	systemctl disable --now avahi-daemon [.socket]	systemctl reports inactive for both units	Removes an unnecessary network-facing local-discovery service
Ubuntu package updates	9 packages shown as upgradable	sudo apt update && sudo apt upgrade -y	Update commands run; final apt list --upgradable re-check recommended as follow-up evidence	Reduces exposure to known, already-patched software issues

Externally Observable TCP Ports — Before vs After

A useful secondary before/after data point is how the externally observed TCP port states changed once UFW was enabled:

Scan	Result on 1,000 default TCP ports
Before hardening (nmap -sV 192.168.56.101, Section 7.1)	All 1,000 ports closed — reset (RST) sent for each probe
After hardening (nmap -sV 192.168.56.101)	All 1,000 ports filtered — no response (silently dropped)

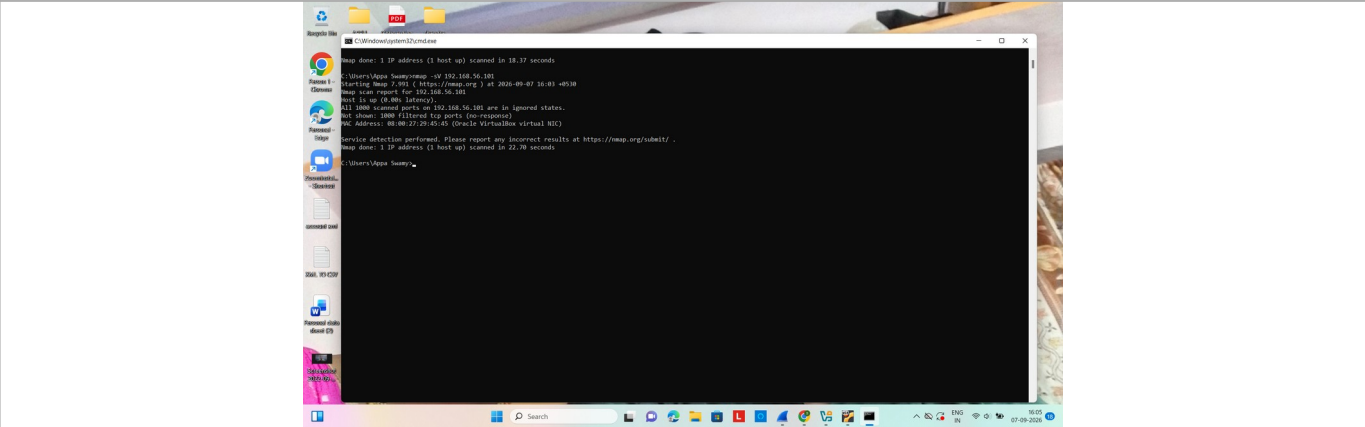


Figure 20. Post-hardening nmap -sV 192.168.56.101 — ports now reported as filtered (no-response) rather than closed (reset).

This does not mean the Ubuntu system has no services at all — it means that, from an external scanner's point of view, UFW is now silently dropping probes to closed ports instead of sending a TCP reset. This is the expected and desired effect of enabling a default-deny firewall.

15. Task 14 — Final Mini Security Assessment

Network Discovery

Host discovery identified three authorized systems on the 192.168.56.0/24 lab segment: the Host-Only gateway (192.168.56.1), a second VirtualBox adapter/VM (192.168.56.100), and the assessment target, the Ubuntu Desktop VM (192.168.56.101). No TCP or UDP services were found open on the target's default scanned port sets; the only notable service-level finding was Avahi/mDNS on UDP 5353, confirmed via host-level evidence.

Traffic Analysis

Wireshark captures on the Windows host demonstrated all five required traffic types — DNS, TCP, UDP, ICMP (via ICMPv6 neighbor discovery), and ARP — all consistent with normal operating system and browser background activity. No packet-level evidence directly correlating Nmap scan traffic against 192.168.56.101 was available in this evidence set; this is flagged as an area for a follow-up, segment-specific capture.

Security Assessment

Five findings were identified and risk-classified, all rated Medium or Low-Medium: potential mDNS/Avahi exposure, an initially inactive firewall, pending package updates, inherent UDP-scan ambiguity, and general attack-surface considerations from running unnecessary services. Each finding is supported by direct command-line or Nmap evidence rather than assumption.

Hardening

Three improvements were implemented: UFW was enabled with a default-deny-incoming policy, Avahi/mDNS was disabled and confirmed inactive at the host level, and pending Ubuntu package updates were reviewed with the update/upgrade commands run.

Verification

Before/after Nmap scans showed a clear, measurable change: default TCP ports moved from closed (reset) to filtered (no-response) after UFW was enabled — direct evidence that the firewall is now actively intercepting probe traffic.

Attack Surface

Enabling the firewall and disabling the unnecessary Avahi service reduced the system's externally observable footprint: probes to the host now receive no response at all rather than an RST confirming the host's presence and port state, and one previously-listening local-discovery service has been removed from the running configuration entirely.

16. Security Recommendations

- 1. Maintain an active host-based firewall such as UFW, with a default-deny-incoming policy.
- 2. Disable unnecessary network services such as mDNS/Avahi where local service discovery is not required.
- 3. Apply operating-system and application security updates regularly.
- 4. Periodically scan authorized systems to identify unexpected exposed services before they become a problem.
- 5. Monitor network traffic and investigate unusual connection patterns, unexpected protocols, or unexpected services, without assuming malicious intent until the evidence supports it.

17. Challenges Faced

- Configuring VirtualBox Host-Only networking correctly so the Ubuntu VM was reachable from the Windows host
- Establishing reliable Windows-to-Ubuntu connectivity for repeated scanning
- Understanding Nmap's UDP open|filtered result and why it cannot be treated as a confirmed open port
- Interpreting Wireshark display filters correctly to isolate DNS, TCP flag, and ARP traffic of interest
- Correlating Nmap scan activity with Wireshark packet captures across two different network interfaces (the isolated lab segment vs. the host's general network)
- Verifying that firewall and service hardening changes had actually taken effect, rather than assuming success from the command output alone

18. Conclusion

Week 3 walked through the complete cybersecurity assessment lifecycle in miniature:

DISCOVER → ENUMERATE → CAPTURE → ANALYZE → ASSESS → HARDEN → VERIFY

Network reconnaissance (host discovery and port scanning) established a clear, evidence-based picture of what was actually reachable on the lab target — notably, that the default scan found no open ports, which is itself a meaningful and accurate result rather than an anticlimax. Service enumeration combined with host-level evidence (ss output) surfaced the one genuinely actionable finding of the assessment: an unnecessary Avahi/mDNS listener. Packet analysis with Wireshark demonstrated the mechanics of DNS, TCP, UDP, ICMP, and ARP traffic in practice, while also surfacing a real gap in the available evidence — a segment-specific Nmap+Wireshark correlation — that is documented honestly rather than glossed over.

The security risk identification and hardening stages closed the loop: three concrete improvements (enabling UFW, disabling Avahi, and reviewing package updates) were implemented and then verified with a second round of scanning, which showed a clear, measurable shift from closed (reset) to filtered (no-response) ports — direct evidence that the hardening had a real, observable effect on the system's external footprint.

Overall, this week reinforced that a security assessment is only as credible as its evidence: distinguishing a confirmed open port from an ambiguous open|filtered state, a host-level listening service from a network-reachable one, and a routine protocol behaviour from a genuine anomaly are the habits that separate a professional assessment from a checklist exercise.

19. References

- Nmap Official Website — <https://nmap.org>
- Nmap Reference Guide — <https://nmap.org/book/man.html>
- Nmap NSE Documentation — <https://nmap.org/book/nse.html>
- Wireshark Official Website — <https://www.wireshark.org>
- Wireshark User's Guide — https://www.wireshark.org/docs/wsug_html_chunked/
- Wireshark Display Filter Reference — <https://www.wireshark.org/docs/dfref/>
- MITRE ATT&CK — <https://attack.mitre.org>
- CVE Program — <https://www.cve.org>
- Ubuntu Security Notices / Update Documentation — <https://ubuntu.com/security>

20. Evidence Index

All screenshots referenced in this report are embedded inline within their corresponding sections. This index summarises each figure for quick reference.

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